

LG-25-081

April 21, 2025

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Limerick Generating Station, Unit 1
Renewed Facility Operating License No. NPF-39
NRC Docket No. 50-352

Subject: LER 2025-001-00 Bypass Leakage on the B Train of Standby Gas Treatment System
HEPA Filters Exceeded Technical Specification Value

In accordance with the requirements of 10 CFR 50.73(a)(2)(i)(B), Limerick Generating Station hereby submits the enclosed Licensee Event Report.

There are no commitments contained in this letter.

If you have any questions, please contact Jordan Rajan at (610) 718-3400.

Respectfully,

Gillin, Michael F. Digitally signed by Gillin, Michael
F.
Date: 2025.04.21 17:39:22 -04'00'

Michael F. Gillin
Vice President – Limerick Generating Station
Constellation Energy Generation, LLC

cc: Administrator Region I, USNRC
USNRC Senior Resident Inspector, Limerick Generating Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Limerick Generating Station Unit 1	☒ 050	2. Docket Number 352	3. Page 1 OF 3
	☐ 052		

4. Title Bypass Leakage on the B Train of Standby Gas Treatment System HEPA Filters Exceeded Technical Specification Value

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☒ 050	Docket Number
02	19	2025	2025	- 001 -	00	04	21	2025	Limerick Unit 2		353
									Facility Name	☐ 052	Docket Number

9. Operating Mode OPCON 1	10. Power Level 100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact Jordan Rajan	Phone Number (Include area code) 610-718-3400
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
A	BH	FLT	A220	Yes					

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On February 19, 2025, at 08:40 the B train of Standby Gas Treatment System (SGTS) was removed from service to test the charcoal adsorber and High-Efficiency Particulate Air (HEPA) filter banks. The test identified the as-found leakage for the upstream and downstream HEPA filter banks exceeded the Technical Specification (TS) leakage allowance. The HEPA filters were replaced, and the system was satisfactorily retested. This condition was reportable per 10 CFR 50.73(a)(2)(i)(B).

The B train of SGTS was previously tested on March 19, 2020, and the as-found leakage was elevated. Since the leakage was close to the TS acceptance criteria, the issue was entered into the corrective action process. The HEPA filters exceeded the leakage criteria because Engineering did not ensure maintenance was scheduled and executed prior to exceeding the TS limit. An action to perform post trending analysis will be added to the Surveillance Test package to ensure the filters do not exceed the TS leakage criteria before the next scheduled performance.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME

Limerick Generating Station Unit 1

**050****052****2. DOCKET NUMBER**

352

3. LER NUMBER**YEAR**

2025

SEQUENTIAL**NUMBER**

001

REV**NO.**

00

NARRATIVE**Unit Condition Prior to the Event**

Unit 1 was in Operational Condition (OPCON) 1 (Power Operation) at approximately 100% power. Unit 2 was in Operational Condition (OPCON) 1 (Power Operation) at approximately 100% power. There were no other structures, systems or components out of service that contributed to this event.

Description of the Event

On February 19, 2025, at 08:40 the B train of Standby Gas Treatment System (SGTS) [E1IS:BH] was removed from service to test the charcoal adsorber and High-Efficiency Particulate Air (HEPA) filter banks. Each SGTS train has two HEPA filter banks and a charcoal adsorber. The upstream HEPA filter bank removes radioactive particulates, while the charcoal adsorber removes gaseous iodine that may be present in the air stream. The downstream HEPA filter bank removes any carbon fines that may become entrained in the air stream after leaving the charcoal adsorber and providing redundant protection against failure of the upstream HEPA filter.

Leakage must be less than 0.05 percent to meet the Technical Specification (TS) limit for in place penetration and leakage testing of the HEPA filter banks. The test identified the as-found leakage for the upstream HEPA filter bank was 0.11 percent and the as-found downstream HEPA filter bank leakage was 0.16 percent. The HEPA filters were replaced, and the system was retested. The system was declared operable on February 20, 2025, at 11:44.

The B train of SGTS was last tested on March 19, 2020. The as-found leakage of the upstream filter bank was 0.044 percent, and the as-found downstream HEPA filter bank was 0.046 percent. Since the leakage was close to the TS acceptance criteria, the issue was entered into the corrective action process. A note was added to the Issue Report (IR 4328138) stating the HEPA filters should be replaced prior to the next performance of the test.

The exact time the B train of SGTS was inoperable cannot be determined. Based on the leakage measured in 2020 and the as-found leakage measured in 2025, the leakage likely existed for greater than the TS acceptance criteria for greater than seven days. Additionally, since the exact date the B train of SGTS became inoperable cannot be determined, there were times that the A train of SGTS was likely inoperable while the B train of SGTS was also inoperable. The action for both SGTS trains being inoperable was not entered. This condition is reportable per 10 CFR 50.73(a)(2)(i)(B).

Analysis of the Event

The HEPA filters failed because Engineering did not ensure maintenance was scheduled and executed prior to exceeding the TS limit. The issue report for the degraded high HEPA filter bank leakage was not flagged as system health issue and the subsequent reviews did not identify the challenge to the TS limit.

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	☐ 052		YEAR 2025	SEQUENTIAL NUMBER 001	REV NO. 00

NARRATIVE**Safety Consequence**

There was no actual safety consequence associated with this event. The HEPA filter banks are conservatively credited with an efficiency of 99.0 percent. The as-found bypass leakage is bounded by the Alternate Source Term (AST) assumed SGTS HEPA filter bank efficiency. The B train of SGTS filter train was available to perform its design basis functions during the period where the upstream and downstream HEPA filter bypass leakage percentages exceeded the Technical Specification limits. This event is not a safety system functional failure.

Cause of the Event

The HEPA filters failed because Engineering did not ensure maintenance was scheduled and executed prior to exceeding the TS limit.

Corrective Actions Completed

The HEPA filters were replaced on the B train of SGTS, and the test was re-performed verifying the in-place penetration and leakage testing of the HEPA filter banks met the TS surveillance requirements.

Corrective Actions Planned

- Review open deficiencies on TS equipment that could result in equipment inoperability to ensure the actions to address equipment deficiencies are properly scheduled.
- Add a task to the Surveillance Test work packages that perform Charcoal/HEPA testing for TS trains to perform post testing trending analysis to ensure that the Charcoal/HEPA filters are projected not to exceed the TS leakage criteria before the next scheduled performance.

Previous Similar Occurrences

There have been no previous similar licensee event reports in the past ten years.

Component Data

System: BH (Emergency/Standby Gas Treatment System)

Component: FLT (Filter)

Component Number:

- 0B-F183 Upstream HEPA filter bank
- 0B-F170 Downstream HEPA filter bank

Manufacturer: A220 (AAF International)